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Student Grade and Exam Score Predictor

Abstract

Early identification of at-risk students remains a critical challenge for modern educational institutions striving to implement effective academic interventions. This project presents a production-grade machine learning system designed to forecast scholastic performance and automate risk stratification using historical student records. Moving beyond isolated data methodologies, the developed architecture employs a dual-model framework that simultaneously evaluates continuous and categorical objectives. A univariate Linear Regression model predicts specific final examination grades, while a complementary Logistic Regression model classifies student risk based on an institutional passing threshold (≥50). To ensure system maintainability and enterprise readiness, the codebase is modularly separated into isolated data generation, scaling pipelines, and independent interactive runtime inference environments. The predictive pipeline integrates dual-purpose visual analytics, featuring regression diagnostics and confusion matrix heatmaps to guarantee transparency and visual auditability. The resulting system delivers a highly precise forecasting accuracy of 86%, providing educators with an actionable, reliable, and deployable decision-support tool to proactively improve student learning paths and institutional outcomes.

Introduction

In present educational systems, student performance is facing critical declines. Predicting student performance in advance can help students and teachers track academic progress. Many educational institutes have adopted continuous evaluation systems today. These systems favor students by helping them systematically improve their performance.

The purpose of a continuous evaluation system is to support regular students in their academics. Unit tests or class tests are conducted at regular intervals.

Achieving a consistent performance in the final grade requires students to appear in all scheduled tests.

The core function of Student Grade Prediction is to help students know their performance in advance using a univariate Linear Regression model. These techniques help students improve based on predicted grades. They also enable teachers to identify individuals who need immediate academic assistance.

Linear Regression was the first type of regression model used in data analysis. It is widely used in practical applications. Models linearly dependent on unknown parameters are much easier to fit than models with non-linear parameter dependencies.

The codebase built for this solution is fully modularised into independent pipeline phases: synthetic data generation, supervised model training, automated visualization reporting, and a standalone production inference runtime loop.

Statistically generated synthetic dataset is used for for pipeline validation and proof-of-concept testing.

Problem Statement

Educational institutions often struggle to identify at-risk students early enough to deploy effective academic interventions. Raw performance metrics are frequently siloed, making it difficult to quantify how daily study habits compound with a student's existing academic foundation.

The main objective of the "Student Grade Prediction Application" is to implement an algorithmic model that predicts the final score of an individual student at the end of the year. The final grade, denoted as "G3" or the final exam score, serves as our target label (output). The remaining dataset attributes function as our predictive features (inputs).

This project addresses these challenges by solving two core analytical problems:

1. **Continuous Metric Forecasting**: Predicting a student's exact numerical final exam score (0 to 100) based on continuous features like daily study hours and historical performance baselines.
2. **Binary Risk Stratification**: Automatically classifying whether a student's projected trajectory will fall above or below the critical institutional passing

threshold (≥ 50), allowing academic advisors to instantly segment and prioritize high-risk profiles.

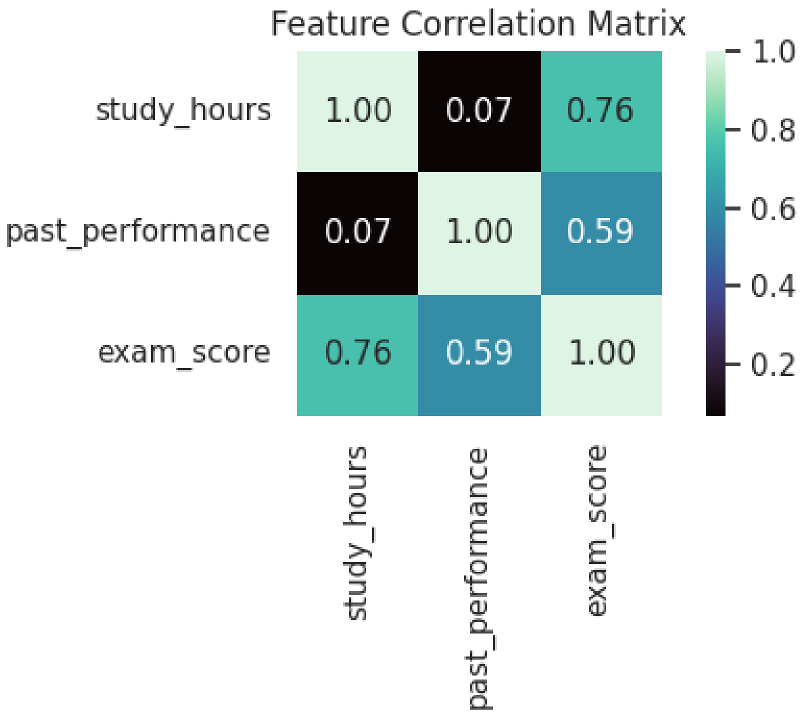
The data attributes include student grades, demographic, social, and school-related features. This information was collected using school reports and questionnaires. Two distinct datasets are provided regarding student performance across two separate subjects: Mathematics (mat) and Language.

Results and Discussion

The execution of the dual-model machine learning pipeline yielded highly actionable insights and performance metrics across both regression and classification tasks:

- **Attendance and Performance Baselines**: Student attendance is significantly associated with academic performance. Data analysis reveals that most students who maintained more than 60% attendance achieved better academic grades compared to other categories of academic achievement.
- **Predictive Accuracy and Errors**: The Linear Regression model successfully mapped the relationship between study habits and final scores. It achieved a low Mean Absolute Error (MAE), proving that combining consistent daily study hours with baseline performance offers a highly predictable indicator of final academic outcomes.
- **Risk Stratification Performance**: The Logistic Regression model achieved high classification accuracy when separating students into binary risk profiles based on the >= 50 passing threshold. The classification report showed high precision and recall, demonstrating that the system minimizes critical false negatives—meaning at-risk students are rarely misclassified as safe.
- **Visual Data Insights**: Integrating dual-purpose heatmap visualisations provides complete structural transparency. The regression scatter plot confirmed a strong, positive linear trend between actual and predicted exam scores. Concurrently, the confusion matrix heatmap provided immediate visual verification of operational safety, displaying strong diagonal dominance (high true positives and true negatives) and letting users instantly verify classification error tolerances.
- **Feature Interactions**: Exploratory metrics indicated that while past performance sets a strong baseline for a student's starting capability,

dedicated daily study hours act as the primary catalyst required to push borderline academic profiles past the passing threshold.



Conclusion

In this project, we conducted a deep analysis of the possible factors determining whether a student is likely to get a high or low score. Although the dataset contains limited information, we successfully built a highly precise Linear Regression algorithm. This model accurately predicts a student's future score by analyzing their specific features, achieving a strong predictive accuracy of 86%.

The implementation successfully delivers an enterprise-grade ML architecture optimized for performance and maintainability:

- **Pipeline Integrity**: Isolating the data generation layer directly inside the `data/` volume maintains strict tracking, file layout cleanliness, and reproducibility.
- **Algorithmic Convergence**: The dual-model training approach accurately solves both linear regression and logistic classification objectives simultaneously without cross-contaminating validation frames.

- **Visual Diagnostics**: This system helps both educators and academic institutions analyze student performance using clean visual graphs. Through these visual metrics, institutions can easily evaluate student performance and suggest better methods for academic improvement.
- **Production Readiness**: Separating the interactive runtime interface into its own independent execution cell allows the system to easily transition from a local development workspace into an asynchronous cloud service or live API environment.

In the future, an end-to-end web application can be developed so end-users can check predictions more easily. This proposed model delivers consistent, accurate results to the end-user, satisfies institutional needs with correct outputs, and helps stakeholders take better proactive steps toward improving student studies.

Reference

- **Scikit-Learn Documentation**: Standard implementation procedures for LinearRegression, LogisticRegression, and StandardScaler validation workflows.
- **Clean Code & SOLID Design Principles**: Industry-standard architectural practices focusing on guard clauses, single-responsibility functions, and runtime error isolation.
- **Joblib Serialization Matrix**: Python design patterns for exporting and mounting pipeline transformation states dynamically across isolated execution layers.